WEARABLE SYSTEM FOR MONITORING POSTURE AND HUMAN MOVEMENTS

MS3 05/05/2026

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Motion Suit

Our project involves developing a suit equipped with sensors that measure body posture and detect anomalies

Data Collection

Personal Data

Name
Gender
Date of Birth
Height
Weight

Health Data

Injury History
Medical Notes

Sensor Data

Body Temperature
Joint Angles

GDPR

Identity Protection

A user's real identity is never directly linked to their biometric data due to UUIDs generated by Keycloak

Local Chatbot

To keep users' data private when asking the chatbot for advice, we run a local LLM instead of an external API

Terms and Conditions

Users must read and accept the Terms and Conditions before creating their account

Research Participation

Researchers only have access to data from users who have authorized data sharing

Right to Erasure

Deleting an account permanently anonymizes all associated records, while raw sensor data is automatically deleted after 90 days

Licenses

MIT Licenses
Allows full use without forcing us to open-source

Frontend: React, Next.js, Three.js
Backend: FastAPI, SQLAlchemy
Hardware: Adafruit BNO055 Drivers

Apache 2.0
Includes explicit patent protection

Identity: Keycloak Server
Database: TimescaleDB
Networking: Python Requests Library

GPL
Using this with our main app forces open-source

Database Adapter: Psycopg2
Raspberry Pi: Lgpio

Specific Licenses
Custom but permissive allowing full integration

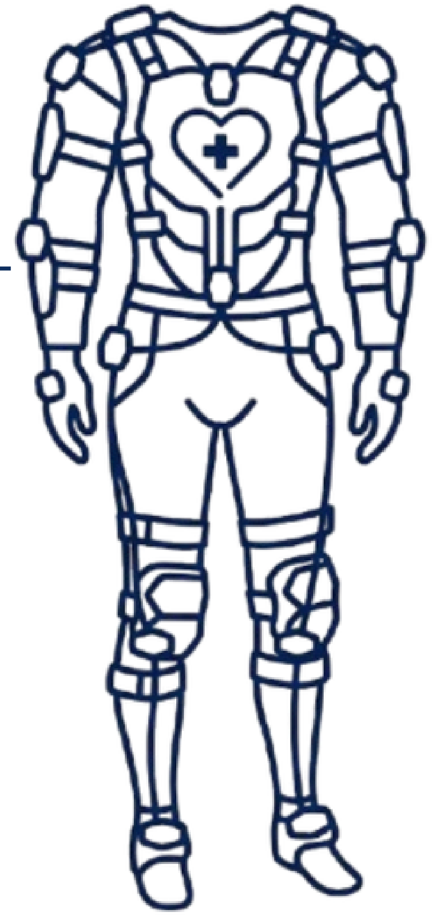
PostgreSQL License: PostgreSQL Database

Licenses

- GPL
- Specific Licenses
- Apache 2.0 Licenses
- MIT Licenses

Our microservices-based architecture allows us to use GPL only at the hardware without forcing our entire project to become open-source

Sector Regulation



ISO 9241

Systems must translate complex data into clear, understandable user feedback

↓

Sensor angles are transformed into an 3D visual interface, allowing users to understand and correct their posture

EU MDR 2017/745

Wearable devices that monitor physical conditions have strict medical rules

↓

We implemented data validation rules to guarantee that only the correct sensor readings are processed

ERS

Must provide reliable feedback to users

↓

We process raw sensor data in our backend before triggering any posture alerts

Terms and Conditions

Conditions of Use

We specify that MotionSuit is only for personal ergonomic awareness, ensuring users don’t misuse the system

Liability Disclaimers

To protect out project from health-related claims, we make it clear that MotionSuit only offers posture advice and is never a substitute for medical treatment

Legal Jurisdiction

Any legal disputes will fall under Portuguese and European Union law

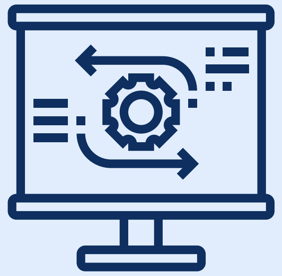
Modification to Terms

We can update these terms with a 30-day email notice to users, giving us the flexibility to adapt our rules as the project evolves

External Dependencies

	Risk identified	Mitigation
Ollama	Discontinuation of support for integration libraries	A modular architecture that allows changing the local AI engine without rewriting the service
Keycloak	Complexity of updating and dependence on OpenID support	Use of standard protocols (OAuth2) to allow migration to other SSO providers
Google Fonts	Build failure in isolated environments or without internet access	Migration to local sources
Google SMTP	Give customers' email addresses to Google	Having our own email server

Technical Debt

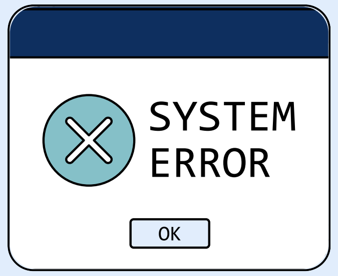


Lack of Automated Testing

We prioritized feature implementation over creating unit and integration tests

Dead Code

We left previous versions of the 3D Skeleton and outdated functions



Raspberry Pi Errors Logs

Hardware logs are stored locally on the Raspberry Pi to avoid overhelming the backend

Refactor Plans

Implement the missing tests to verify that all features function as intented

Identify and remove unused files and functions to clean up the code

Implement a structured log-syncing system to push critical hardware errors to backeng

Fragility of Critical Components

Bittle Component	Keycloak-Postgres Sync	NextAuth.js (v5 Beta)
Why fragile?	Onboarding flow relies on exact mapping between Keycloak tokens and the users table schema	Currently using an unstable (beta) version where internal APIs change frequently
Impact	Total block for new user registrations	Critical login failures or broken builds following automatic package updates
Mitigation	Implementation of a robust Service Layer to validate data before insertion and allow for schema flexibility	Strict version pinning in package.json and creating integration tests for the auth flow

Fragility of Critical Components

Bittle Component	REBA/ROSA Calculation Logic	Remote Build Dependencies
Why fragile?	Complex mathematical algorithms that expect a rigid JSON format from the sensors	The frontend build process requires an active internet connection to fetch external assets
Impact	Incorrect ergonomic risk scores or API crashes if a sensor sends a null value	Build failure in isolated environments, preventing the system from being deployed offline
Mitigation	Implement Strict Schema Validation and unit tests for edge cases	Migrate all build-time assets to local storage within the project's public/ directory

SWOT

	POSITIVES TO BE EXPLOITED	NEGATIVES TO BE MINIMIZED
INTERNAL TO ORG	**Strengths** • High-fidelity Data Architecture • Edge-Ready & Modular Design • Secure Infrastructure	**Weaknesses** • Hardware Scaling Costs • Supply Chain Vulnerability
EXTERNAL TO ORG	**Opportunities** • Corporate ESG & Insurance Trends • Predictive AI Integration • Telehealth Expansion	**Threats** • Markerless Computer Vision • Worker Privacy Pushback • Strict Medical Regulations

TOWS - Strengths + Opportunities

S2: Edge-Ready Design
+
O1: Corporate ESG & Insurance Trends

Raspberry Pi processes data locally without needing constant cloud connection → We can sell it to production lines where the Wi-Fi is restricted to help them lower workplace injury insurance

S1: High Fidelity Data Architecture
+
O2: Predictive AI

We can use our telemetry data to feed directly into a ML model → The data is clean, we can quickly deploy new models (e.g., real-time feedback to correct posture on the fly)

TOWS - Strengths + Threats

S3: Secure Infrastructure
+
T1: Worker Privacy Pushback

All user data is encrypted, anonymized, and secured by RBAC mechanisms → All data is controlled by the user, they can use the application without consenting to share it with researchers. There is no corporate surveillance whatsoever

S1: High Fidelity Data Architecture
+
T2: Markerless Computer Vision

Cameras cannot capture all of the movements if the workers are hidden behind boxes or machinery → The wearable suit maintain perfect motion tracking, independently of where the worker is

TOWS - Weaknesses + Opportunities

W1: Hardware Scaling Costs
+
O1: Corporate ESG & Insurance Trends

B2b Pilots first (Bosch) → Using corporate wellness and safety budgets to fund the initial investment

TOWS - Weaknesses + Threats

W2: Supply Chain Vulnerability
+
T3: Strict Medical Regulations

We must avoid getting trapped in years of expensive FDA/MDR medical certification while fighting possible hardware shortages

The suit should be marketed as a “workplace wellness and sports performance” tool initially, avoiding the “medical device” label

We should only pivot the focus when the supply chain and capital are secure

PESTEL - Political Factor

Data Sovereignty and GDPR Compliance

Macro factor: Rigorous EU and Portuguese policies on health data privacy

MotionSuit Reality: Local AI processing with Ollama and identity management via Keycloak

Context: Our choice for a local architecture ensures biometric data remains within the user's environment, meeting EU "Data Sovereignty" goals

EU AI Act Compliance

Macro factor: New European legal framework for AI systems in workplace and health contexts

MotionSuit Reality: The JARVIS assistant analyzes session data and joint angles

Context: We must position JARVIS as an advisory tool to avoid the "High-Risk Medical Device" classification under the upcoming AI Act

PESTEL - Economic Factor

Cloud Infrastructure and Currency Risk

Macro factor: High resource demands and USD/EUR exchange rate volatility in AWS

MotionSuit Reality: Docker-based, Cloud-Agnostic architecture

Context: If AWS prices spike by 30%, we can easily migrate to European providers (Hetzner/OVH) to keep costs in Euros and maintain margins

Local AI: The "Zero-Token" Strategy

Macro factor: High and unpredictable pay-per-token costs of external APIs

MotionSuit Reality: Local execution of Ollama (Llama 3) and Whisper

Context: By hosting our own AI, we are immune to API price inflation. Our AI cost is fixed and tied only to our server's energy/maintenance

PESTEL - Social Factor

Digital Literacy and Accessibility

Macro factor: Wide gap in digital skills between younger tech-natives and older industrial workers

MotionSuit Reality: Features like Text-to-Speech, Colorblind Mode, and haptic alerts

Context: By reducing screen dependency, we ensure the suit is accessible to users regardless of their digital literacy level

Generational Trends and Gamification

Macro factor: Gen Z and Millennials embrace "Quantified Self" trends and fitness tracking

MotionSuit Reality: Gamification Service with XP, Streaks and Badges

Context: Using game mechanics to drive engagement among younger workers, while keeping JARVIS simple for older generations

PESTEL - Technological Factor

Tech Maturity and Stability

Macro factor: Need for stable, scalable stacks capable of handling high-frequency sensor time-series data

MotionSuit Reality: Core stack built on FastAPI, PostgreSQL, and TimescaleDB for efficient data compression and continuous aggregates

Context: Isolate experimental components (like local LLMs) in Docker containers. This allows for easy AI model swaps without breaking the stable core system

Hardware Obsolescence vs Software Agility

Macro factor: Rapid evolution of IoT sensors and microcontrollers (Raspberry Pi)

MotionSuit Reality: Decoupled architecture using Docker and standardized JSON payloads

Context: While hardware (Suit V1.0) may become obsolete, our backend remains future-proof. We can integrate "Suit V2.0" (5G/NB-IoT) without changing the core API

PESTEL - Environmental Factor

Data Retention and "Green Storage"

Macro factor: Environmental cost of hosting "dark data" in cloud storage

MotionSuit Reality: Automated compression and retention policies in TimescaleDB

Context: Reduce digital waste by compressing 30-day-old data, significantly lowering the energy required for long-term storage

Smart Inactivity and Batch Processing

Macro factor: High carbon emissions from persistent network connections and idle CPU usage

MotionSuit Reality: Use of batch processing (500 samples per sync) and inactivity detection triggers

Context: Optimize power by shutting down high-frequency polling during inactive periods, reducing both battery drain and server load

PESTEL - Legal Factor

Algorithmic Management Restrictions

Macro factor: New labor directives prohibiting the use of AI/IoT for automated worker surveillance

MotionSuit Reality: Backend detects inactivity periods and movement reminders

Future impact: We may be legally forced to remove "inactivity tracking" to focus strictly on ergonomics, ensuring the suit is not used as a tool for labor monitoring

EU AI Act Compliance

Macro factor: New EU regulation classifying AI systems based on risk levels

MotionSuit Reality: JARVIS provides postural advice based on biometric and health data

Future impact: Risk of being classified as a "High-Risk AI System," requiring expensive conformity assessments and human-in-the-loop supervision

CALENDAR

May 5
MS3: Legal requirements, technical and business risks

T-12 — Done
Compare data from different users

T-14 — Done
Implement chatbot

May 19
MS4: System component validation

T-13 — In Progress
Integration with a smartwatch

T-3 — In Progress
View real-time metrics on the dashboard

T-7 — To do
Calibrate the suit

IEETA deti

THANK YOU!

https://pei-motionsuit.github.io/microsite/

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